

MIC regression

2025-11-06

```
library(tidyverse)
library(cowplot)
library(readxl)
library(viridisLite)

# install_github("lu-harr/iddoPal")
library(iddoPal)
# install_github("lu-harr/looseVis")
library(looseVis)

dat <- read_excel("GAS_SR_Raw_50_90_NOS_formatted_20251106.xlsx")
```

Here are some quick summary stats:

```
# number of isolates
sum(dat$count)
```

```
## [1] 39737
```

```
# number of isolates per country
dat %>%
  group_by(country) %>%
  summarise(n = sum(count)) %>%
  arrange(desc(n))
```

```
## # A tibble: 47 x 2
##   country      n
##   <chr>      <dbl>
## 1 Japan      5984
## 2 Spain      5449
## 3 Serbia     5282
## 4 USA        3968
## 5 Brazil     2710
## 6 Italy       2351
## 7 Worldwide  1485
## 8 Canada     1271
## 9 Austria    1084
## 10 Germany    836
## # i 37 more rows
```

```
# number of isolates per who region
dat %>%
  group_by(who) %>%
  summarise(n = sum(count)) %>%
  arrange(desc(n))
```

```
## # A tibble: 7 x 2
##   who      n
```

```
##    <dbl> <dbl>
## 1      4 20814
## 2      2  9024
## 3      6  7040
## 4     NA  2174
## 5      3   331
## 6      1   270
## 7      5    84

# combined mic
mic <- dat %>%
  mutate(nos_90_comb = ifelse(is.na(micnos_1), mic90_1, micnos_1))

which(is.na(mic$nos_90_comb))

## integer(0)

nos_vec <- rep(mic$micnos_1, mic$count)
median(nos_vec, na.rm=TRUE)

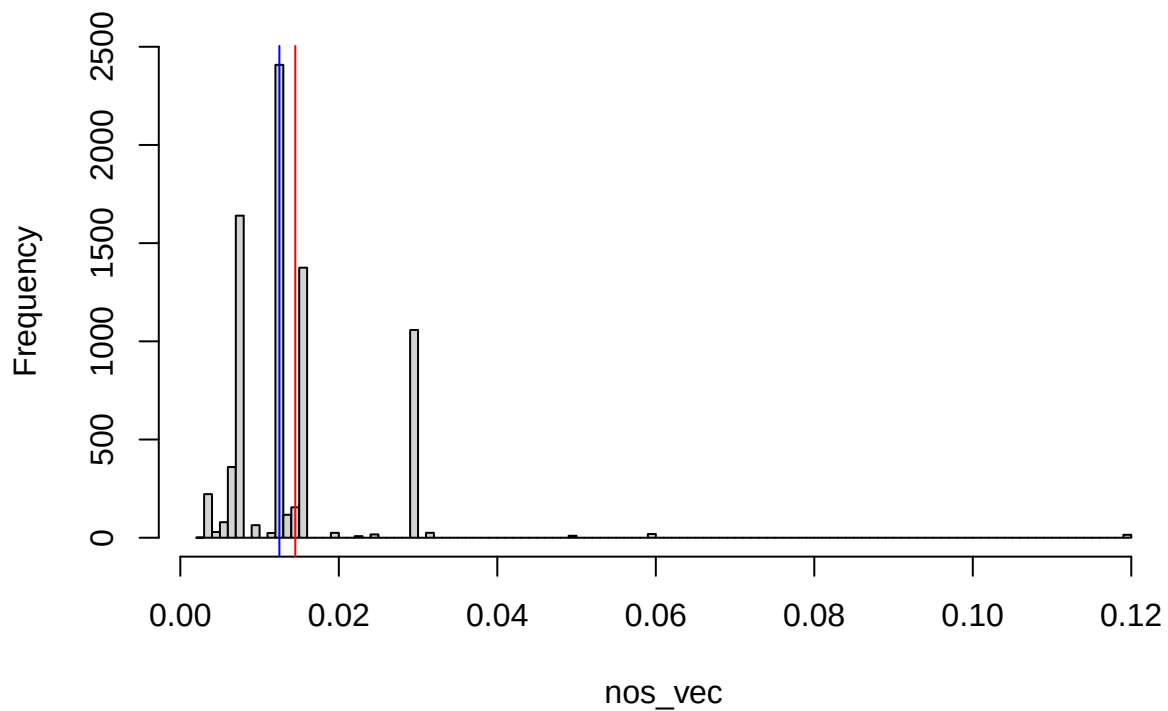
## [1] 0.0125

mean(nos_vec, na.rm=TRUE)

## [1] 0.0145049

hist(nos_vec, breaks=100)
abline(v = median(nos_vec, na.rm=TRUE), col="blue")
abline(v = mean(nos_vec, na.rm=TRUE), col="red")
```

Histogram of nos_vec



```
# represent entries with multiple isolates multiple times:
nos_90_comb_vec <- rep(mic$nos_90_comb, mic$count)
length(nos_90_comb_vec)
```

```
## [1] 39737
median(nos_90_comb_vec)

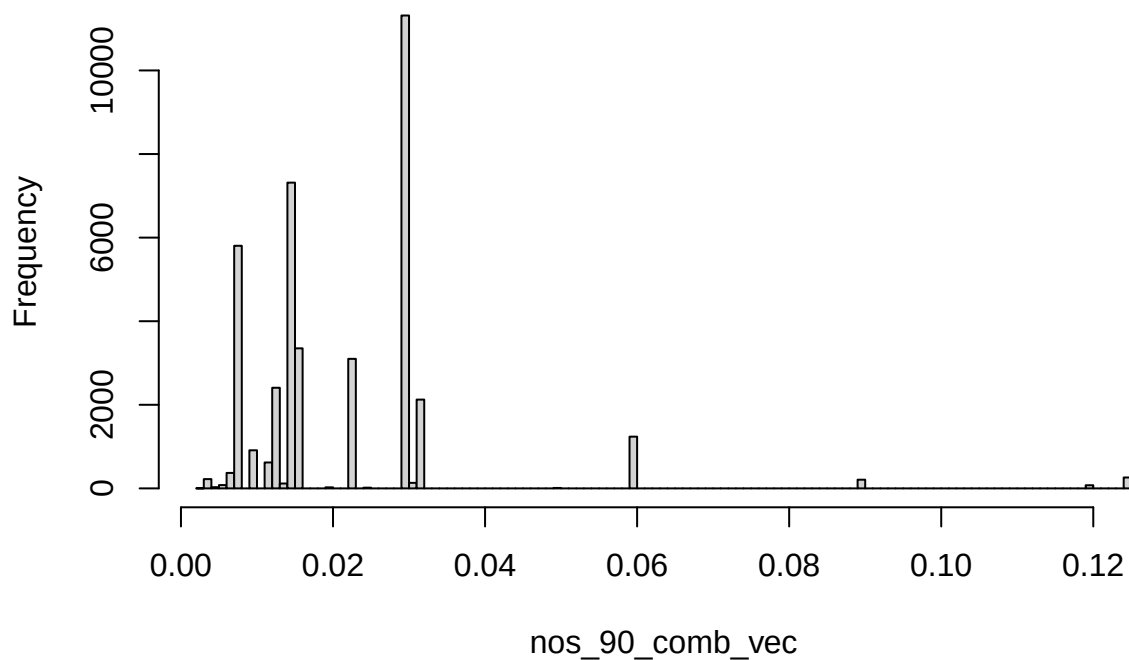
## [1] 0.016
quantile(nos_90_comb_vec, probs = c(0.1, 0.5, 0.9))

## 10% 50% 90%
## 0.008 0.016 0.031
range(nos_90_comb_vec)

## [1] 0.002 0.125
summary(nos_90_comb_vec)

## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.00200 0.01250 0.01600 0.02217 0.03000 0.12500
hist(nos_90_comb_vec, breaks = 100)
```

Histogram of nos_90_comb_vec



And now onto the regression ...

```
who_df <- matrix(c(c("1", "Africa"),
                    c("2", "Americas"),
                    c("3", "Eastern Mediterranean"),
                    c("4", "Europe"),
                    c("5", "Southeast Asia"),
                    c("6", "Western Pacific")),
                 ncol = 2, byrow = TRUE) %>%
data.frame() %>%
```

```

setNames(c("nume", "name"))

who_lst = who_df$name
names(who_lst) = who_df$nume

# and join to data:
dat <- dat %>%
  mutate(who = as.factor(who)) %>%
  left_join(who_df, by = join_by(who == nume)) %>%
  mutate(who_name = ifelse(is.na(name), "Not assigned", name)) %>%
  # need to put both MIC_nos and MIC_90 in one column
  mutate(mic_mix = case_when(!is.na(micnos_1) ~ micnos_1,
                             !is.na(mic90_1) ~ mic90_1,
                             TRUE ~ NA),
         weight_mic = 1 / mic_mix)

```

Populate names of WHO regions:

```

mod_unweighted = lm(mic_mix ~ isolate_yr, dat)
summary(mod_unweighted)

```

Variance-weighted least squares

```

##
## Call:
## lm(formula = mic_mix ~ isolate_yr, data = dat)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.019313 -0.009108 -0.005491  0.000399  0.104851
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -1.165e-01  1.974e-01  -0.590   0.555
## isolate_yr   6.850e-05  9.872e-05   0.694   0.488
##
## Residual standard error: 0.02038 on 275 degrees of freedom
## Multiple R-squared:  0.001748, Adjusted R-squared: -0.001882
## F-statistic: 0.4815 on 1 and 275 DF, p-value: 0.4883

```

```

# setting weight to mean absolute residuals
wts <- data.frame(isolate_yr = dat$isolate_yr,
                  resid = mod_unweighted$residuals) %>%
  group_by(isolate_yr) %>%
  summarise(wt = median(abs(resid)))

dat <- dat %>%
  left_join(wts)

```

```

## Joining with `by = join_by(isolate_yr)`

```

```

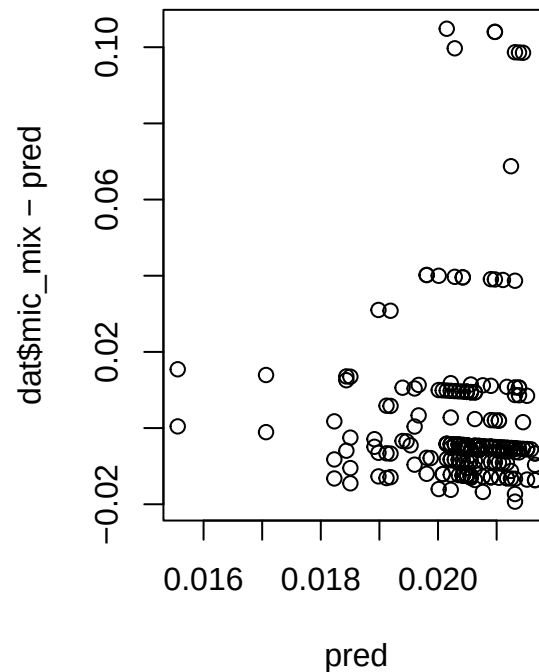
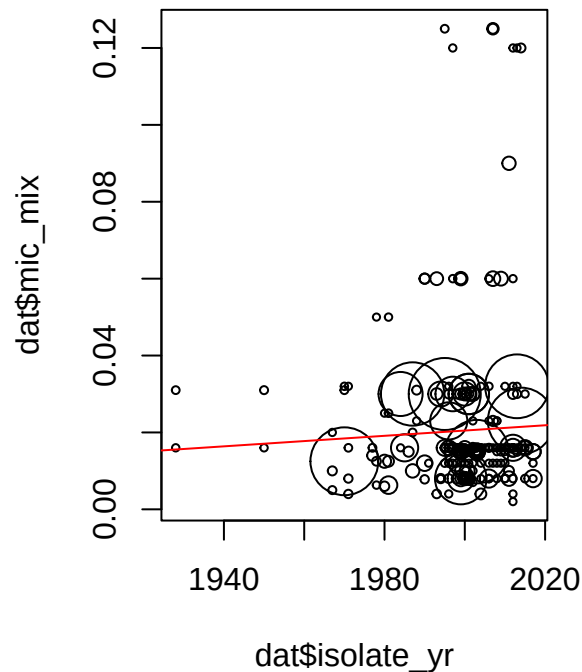
# a quick summary:
par(mfrow = c(1,2))
pred <- predict(mod_unweighted, dat)
plot(dat$isolate_yr, dat$mic_mix, cex = scale_cex(dat$count, max_cex = 5), main = "(unweighted) ordinary

```

```
abline(mod_unweighted, col="red")
plot(pred, dat$mic_mix - pred, main = "Residuals over fitted")
```

(unweighted) ordinary least squares

Residuals over fitted

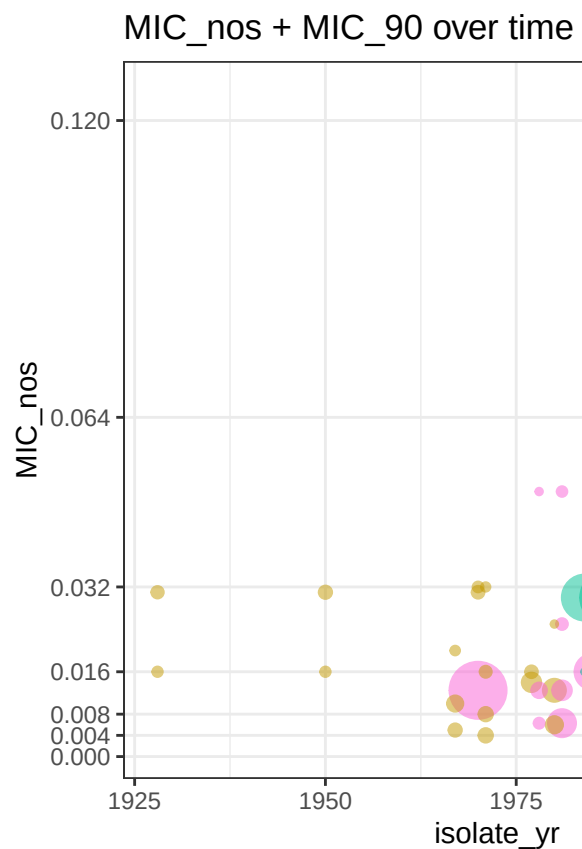


now apply weights to mic_mix

```
yticks <- c(0, 0.004, 0.008, 0.016, 0.032, 0.064, 0.12)

p <- ggplot(dat %>%
  filter(!is.na(mic_mix)) %>%
  # put smaller points in front:
  arrange(desc(count)),
  aes(x = isolate_yr, y = mic_mix)) +
  geom_point(aes(col = who_name,
    size = count), alpha = 0.5) +
  # pop a line on there
  # geom_smooth(method = "lm", se = FALSE, col = "darkgrey") +
  scale_color_discrete(name = "WHO Region") +
  scale_size_continuous(range = c(1,10), name = "Number of isolates") +
  # set specific ticks on y axis: (could transform this too if we think we need to ...)
  scale_y_continuous(breaks = yticks, minor_breaks = NULL, name = "MIC_nos") +
  labs(title = "MIC_nos + MIC_90 over time (points not grouped)") +
  theme_bw()
```

p



Here's the bubble plot with both the MIC_nos and the MIC_90s:

```
gap <- 1

# cobble something together ...
pal <- c(rev(viridis(4)),
        pals$loosepal[3],
        iddo_palettes$iddo[1],
        "orange") %>%
  unname()

# model weighted with weights from above
mod_weighted <- lm(mic_mix ~ isolate_yr, dat, weights = wt)
summary(mod_weighted)
```

Now here's a bubble plot with regression:

```
##
## Call:
## lm(formula = mic_mix ~ isolate_yr, data = dat, weights = wt)
##
## Weighted Residuals:
##      Min       1Q   Median       3Q      Max
## -0.0022383 -0.0011114 -0.0006127 -0.0000777  0.0157608
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.2025994  0.2272423  -0.892   0.373
```

```
## isolate_yr    0.0001125  0.0001137   0.990    0.323
##
## Residual standard error: 0.00219 on 275 degrees of freedom
## Multiple R-squared:  0.00355,    Adjusted R-squared:  -7.306e-05
## F-statistic: 0.9798 on 1 and 275 DF,  p-value: 0.3231

confint(mod_weighted)

##                2.5 %          97.5 %
## (Intercept) -0.6499550168 0.2447561606
## isolate_yr  -0.0001112752 0.0003363511

tmp <- predict(mod_weighted, data.frame(isolate_yr = 1928:2017), se.fit = TRUE)

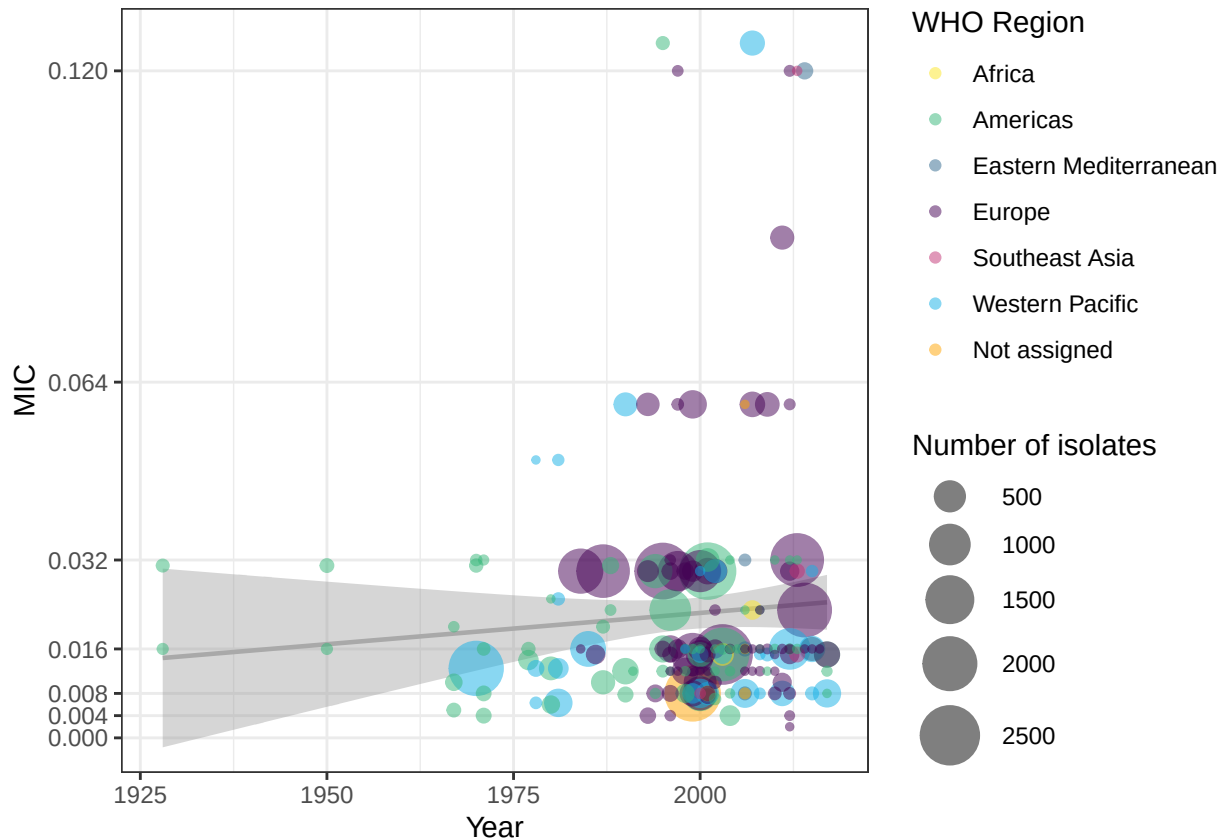
p <- ggplot(dat %>%
  filter(!is.na(mic_mix)) %>%
  group_by(isolate_yr, mic_mix, who_name) %>%
  summarise(count = sum(count)) %>%
  mutate(who_name = factor(who_name, levels = c("Africa",
                                                "Americas",
                                                "Eastern Mediterranean",
                                                "Europe",
                                                "Southeast Asia",
                                                "Western Pacific",
                                                "Not assigned")))) %>%

  # put smaller points in front:
  arrange(desc(count)),
  aes(x = isolate_yr, y = mic_mix)) +
coord_cartesian(ylim = c(0, 0.125)) +
# pop a line on there
geom_smooth(data = dat %>% filter(!is.na(mic_mix)),
  #mapping = aes(weight = count),
  #mapping = aes(weight = weight_mic),
  mapping = aes(weight = wt),
  method = "lm", se = TRUE, col = "darkgrey", linewidth = 0.8) +
#geom_smooth() +
geom_point(aes(col = who_name,
  size = count), alpha = 0.5) +
scale_color_discrete(name = "WHO Region", type = pal) +
# scale_size_continuous(range = c(1,6), name = "Number of isolates",
#                               trans="log10") +
scale_size_continuous(range = c(1,10), name = "Number of isolates") +
# set specific ticks on y axis: (could transform this too if we think we need to ...)
scale_y_continuous(breaks = yticks, minor_breaks = NULL, name = "MIC#",
  #limits = c(0, 0.125)
) +
scale_x_continuous(limits = c(1928 - gap, 2017 + gap)) +
#labs(title = "MIC_nos + MIC_90 over time (points grouped, variance-weighted LS)") +
labs(x = "Year", ylab = "MIC") +
theme_bw()

## `summarise()` has grouped output by 'isolate_yr', 'mic_mix'. You can override
## using the `.groups` argument.
```

p

```
## `geom_smooth()` using formula = 'y ~ x'
```



```
ggsave("figs/mic_mixed_bubbles.png",
        height = 6, width = 9, scale = 0.8)
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

```
# would love to transform the y axis but if i can't extract the ribbon from
# geom_smooth then I can't tell it not to freak out at zero ..
```

```
# sad - doesn't look like coord_cartesian will work with axis breaks
# https://github.com/YuLab-SMU/ggbreak/issues/59
library(ggbreak)
```

Here's the same plot with some breaks in the x axis:

```
## ggbreak v0.1.6 Learn more at https://yulab-smu.top/
```

```
## If you use ggbreak in published research, please cite the following
```

```
## paper:
```

```
##
```

```
## S Xu, M Chen, T Feng, L Zhan, L Zhou, G Yu. Use ggbreak to effectively
```

```
## utilize plotting space to deal with large datasets and outliers.
```

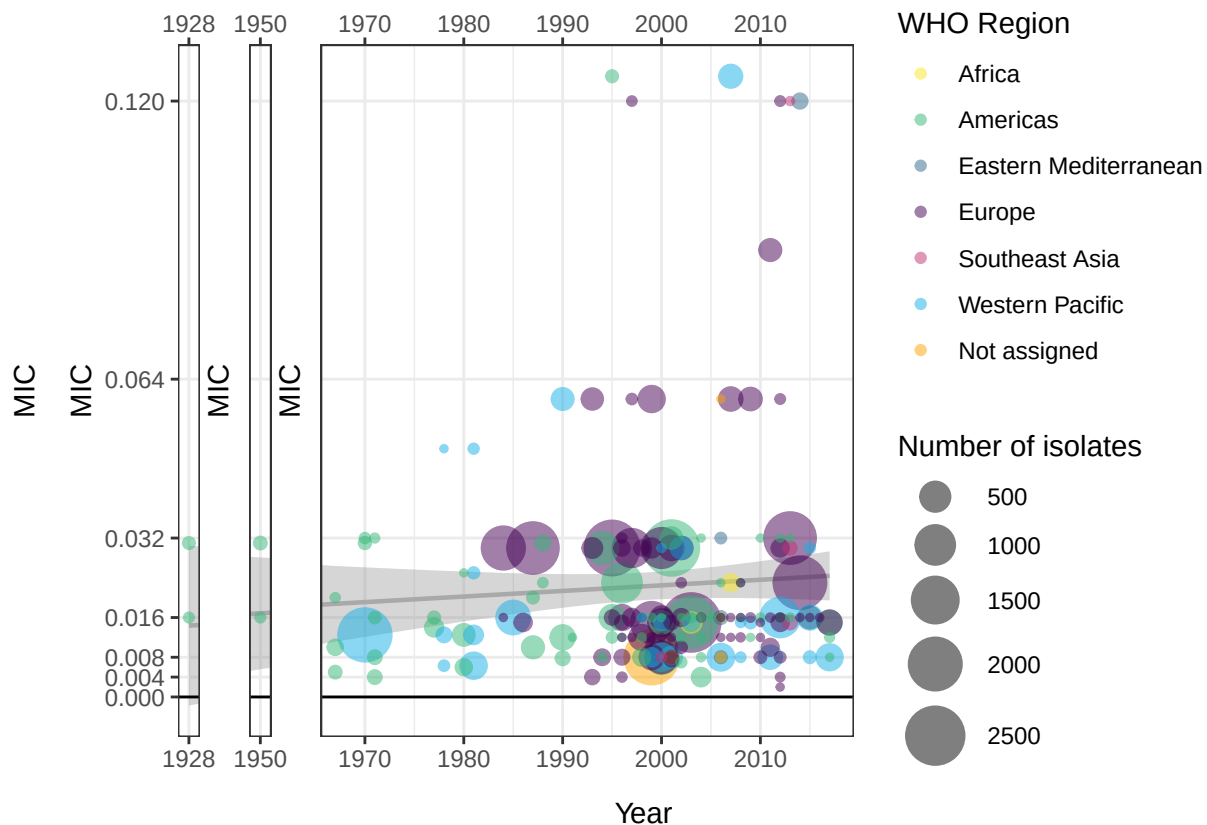
```
## Frontiers in Genetics. 2021, 12:774846. doi: 10.3389/fgene.2021.774846
```



```
p + scale_x_break(c(1928 + gap, 1950 - gap)) +
  coord_cartesian(ylim = c(0, 0.125)) +
  scale_x_break(c(1950 + gap, 1968)) +
  scale_x_continuous(breaks = c(1928, 1950, seq(1960, 2021, 10)),
                    limits = c(1928 - gap, 2016 + gap)) +
  geom_hline(aes(yintercept = 0)) +
  xlab("Year") + ylab("MIC")
```

```
## Coordinate system already present. Adding new coordinate system, which will
## replace the existing one.
```

```
## Scale for x is already present.
## Adding another scale for x, which will replace the existing scale.
## `geom_smooth()` using formula = 'y ~ x'
## `geom_smooth()` using formula = 'y ~ x'
## `geom_smooth()` using formula = 'y ~ x'
## `geom_smooth()` using formula = 'y ~ x'
## `geom_smooth()` using formula = 'y ~ x'
## `geom_smooth()` using formula = 'y ~ x'
```



```
ggsave("figs/mic_mixed_bubbles_with_breaks.png",
       height = 6, width = 9, scale = 0.8)
# make colours viridis plus an orange maybe?
```

Now here's a plot with the y-axis logged but *not* the regression: (so regression's still on a linear scale)

```

Y_LOWER <- 0.0018
X_LOWER <- 1925
mod_weighted <- lm(mic_mix ~ isolate_yr, dat, weights = wt)
confint(mod_weighted)

##                2.5 %          97.5 %
## (Intercept) -0.6499550168 0.2447561606
## isolate_yr  -0.0001112752 0.0003363511

ci <- predict(lm(mic_mix~isolate_yr, dat, weights = wt),
              newdata = data.frame(isolate_yr = seq(X_LOWER,
                                                    max(dat$isolate_yr) + 2)),
              interval = "confidence") %>%
  as.data.frame() %>%
  mutate(lwr = ifelse(lwr < Y_LOWER, Y_LOWER, lwr),
         isolate_yr = seq(X_LOWER, max(dat$isolate_yr) + 2))

# okay: need to transform y-axis and truncate
gap <- 1

# cobble something together ...
pal <- c(rev(viridis(4)),
        pals$loosepal[3],
        iddo_palettes$iddo[1],
        "orange") %>%
  unname()

yticks <- c(0, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.12)

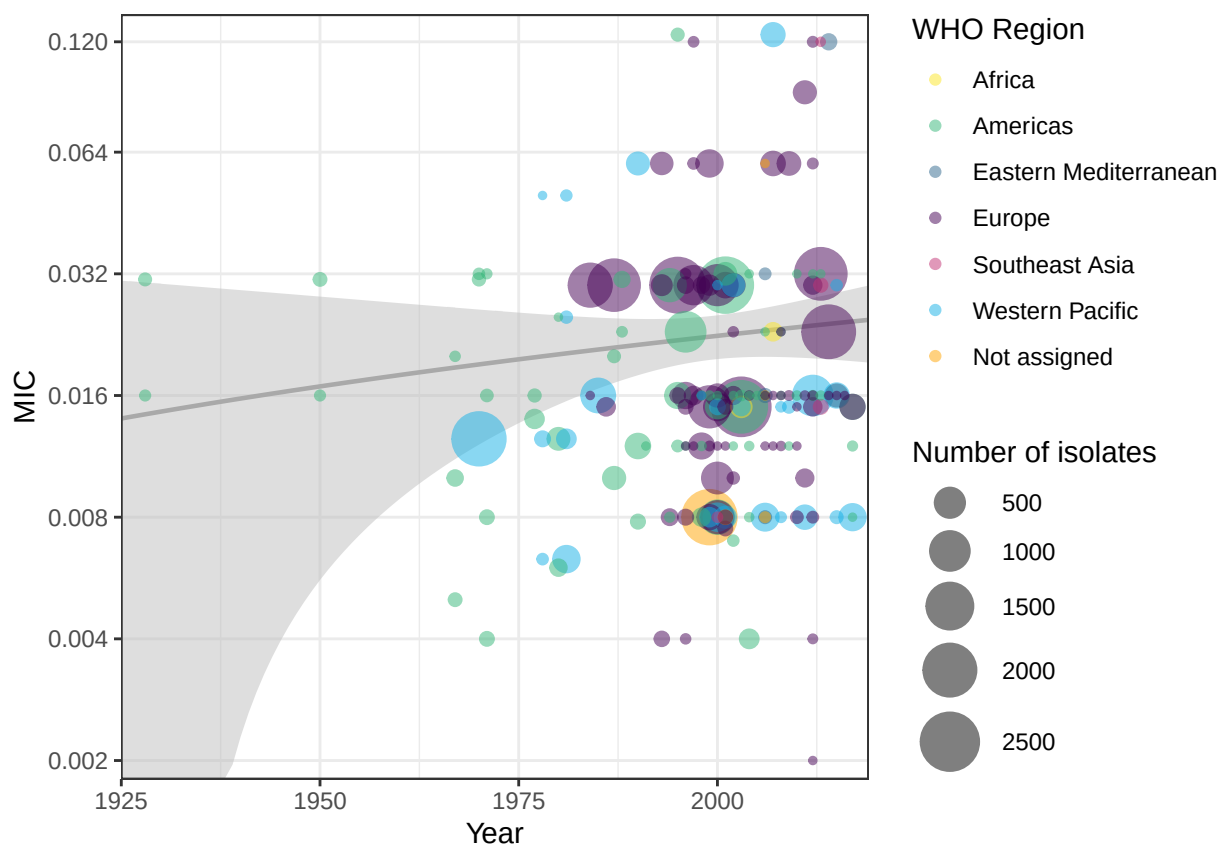
library(ggbreak)
p <- ggplot(dat %>%
  filter(!is.na(mic_mix)) %>%
  group_by(isolate_yr, mic_mix, who_name) %>%
  summarise(count = sum(count)) %>%
  mutate(who_name = factor(who_name, levels = c("Africa",
                                                "Americas",
                                                "Eastern Mediterranean",
                                                "Europe",
                                                "Southeast Asia",
                                                "Western Pacific",
                                                "Not assigned")))) %>%
  arrange(desc(count))) +
  geom_ribbon(data = ci, aes(ymin = lwr, ymax = upr, x = isolate_yr),
            alpha = 0.5, fill = "grey", colour = NA) +
  geom_line(data = ci, aes(x = isolate_yr, y = fit),
            colour = "darkgrey", linewidth = 0.8) +
  geom_point(aes(x = isolate_yr, y = mic_mix,
                 col = who_name,
                 size = count), alpha = 0.5) +
  scale_color_discrete(name = "WHO Region", type = pal) +
  scale_size_continuous(range = c(1,10), name = "Number of isolates") +
  scale_y_continuous(breaks = yticks, minor_breaks = NULL, name = "MIC") +
  scale_x_continuous(limits = c(X_LOWER, 2017 + 2)) +
  labs(x = "Year", ylab = "MIC") +

```

```
coord_trans(y = "log2", ylim = c(Y_LOWER, 0.14), expand = FALSE) +
theme_bw()
```

```
## `summarise()` has grouped output by 'isolate_yr', 'mic_mix'. You can override
## using the `.groups` argument.
```

```
p
```



```
ggsave("figs/mic_mixed_bubbles_logged.png",
        height = 6, width = 9, scale = 0.8)
```

The bounds of the confidence interval for mean MIC get a bit weird at low MICs (early years). What scale are we interpreting change in mean MIC on? Probably on the log scale also ... for this reason, we do the regression on the log scale below.

Alternatively, do the regression on the log scale and re-exp the labels This is the model + visualisation we include in the manuscript. Given tested MICs are doubling, we carry out regression on the log2 scale.

```
# apply log to the mics:
dat <- dat %>%
  mutate(mic_logged = log2(mic_mix))

# fit an unweighted model to determine weights
mod_unweighted = lm(mic_logged ~ isolate_yr, dat)
summary(mod_unweighted)
```

```
##
## Call:
```

```

## lm(formula = mic_logged ~ isolate_yr, data = dat)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.99224 -0.40633 -0.07964  0.04772  2.98163
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -6.9308585   9.0673854  -0.764   0.445
## isolate_yr   0.0004758   0.0045351   0.105   0.917
##
## Residual standard error: 0.9363 on 275 degrees of freedom
## Multiple R-squared:  4.003e-05, Adjusted R-squared:  -0.003596
## F-statistic: 0.01101 on 1 and 275 DF,  p-value: 0.9165

# setting weight to mean absolute residuals
wts <- data.frame(isolate_yr = dat$isolate_yr,
                  resid = mod_unweighted$residuals) %>%
  group_by(isolate_yr) %>%
  summarise(wt = median(abs(resid)))

dat <- dat %>%
  left_join(wts)

## Joining with `by = join_by(isolate_yr, wt)`

mod_weighted <- lm(mic_logged ~ isolate_yr, dat, weights = wt)
confint(mod_weighted)

##              2.5 %      97.5 %
## (Intercept) -29.453208006  9.89141900
## isolate_yr   -0.007913285  0.01177093

summary(mod_weighted)

##
## Call:
## lm(formula = mic_logged ~ isolate_yr, data = dat, weights = wt)
##
## Weighted Residuals:
##      Min       1Q   Median       3Q      Max
## -0.25675 -0.05960 -0.01067  0.00858  0.45078
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -9.780895   9.992906  -0.979   0.329
## isolate_yr   0.001929   0.004999   0.386   0.700
##
## Residual standard error: 0.09629 on 275 degrees of freedom
## Multiple R-squared:  0.000541, Adjusted R-squared:  -0.003093
## F-statistic: 0.1488 on 1 and 275 DF,  p-value: 0.6999

# this is slope on log-scale:
coefficients(mod_weighted)[2]

## isolate_yr
## 0.001928824

```

```
2**(coefficients(mod_weighted)[2])
```

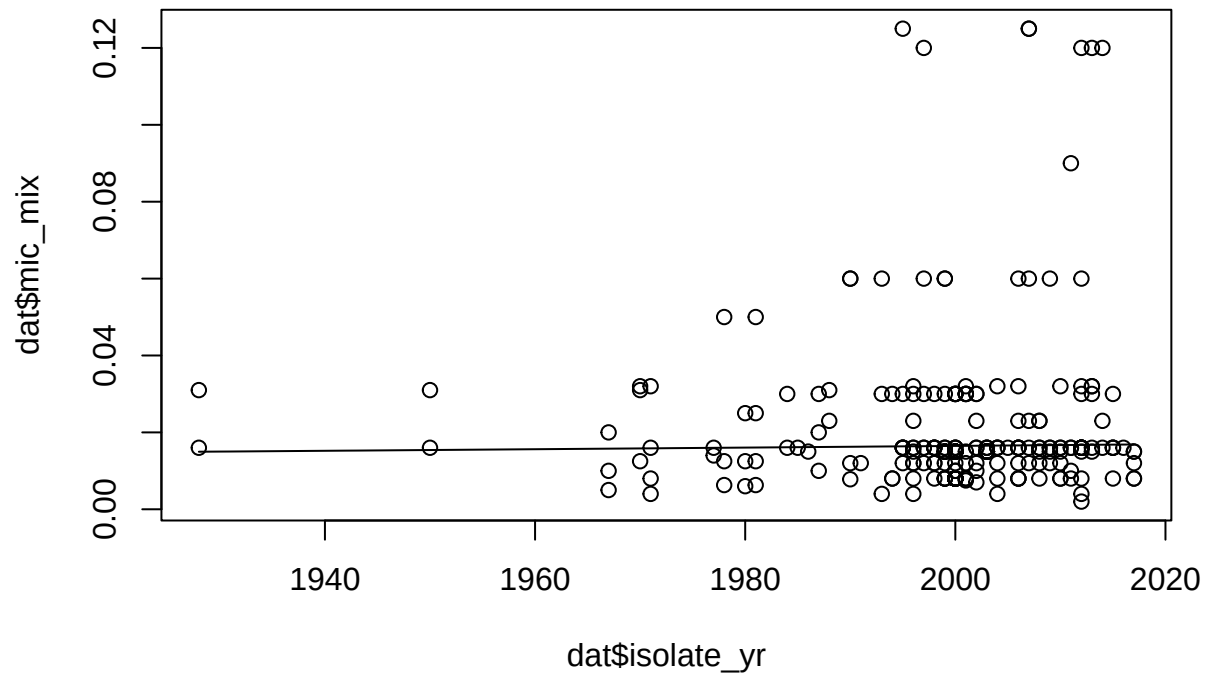
```
## isolate_yr
## 1.001338
```

```
2**confint(mod_weighted)[2,]
```

```
## 2.5 % 97.5 %
## 0.9945299 1.0081924
```

```
# Here's what that looks like unlogged:
```

```
plot(dat$isolate_yr, dat$mic_mix)
yyy = seq(min(dat$isolate_yr), max(dat$isolate_yr))
lines(yyy, 2**predict(mod_weighted, data.frame(isolate_yr = yyy)))
```



```
# (pretty flat)
```

```
Y_LOWER <- log2(0.0018)
X_LOWER <- 1925
```

```
ci <- predict(lm(mic_logged ~ isolate_yr, dat, weights = wt),
              newdata = data.frame(isolate_yr = seq(X_LOWER,
                                                    max(dat$isolate_yr) + 2)),
              interval = "confidence") %>%
  as.data.frame() %>%
  mutate(lwr = ifelse(lwr < Y_LOWER, Y_LOWER, lwr),
         isolate_yr = seq(X_LOWER, max(dat$isolate_yr) + 2))
```

```
gap <- 1
```

```
# cobble something together ...
```

```
pal <- c(rev(viridis(4)),
        pals$loosepal[3],
```

```

    iddo_palettes$iddo[1],
    "orange") %>%
  unname()

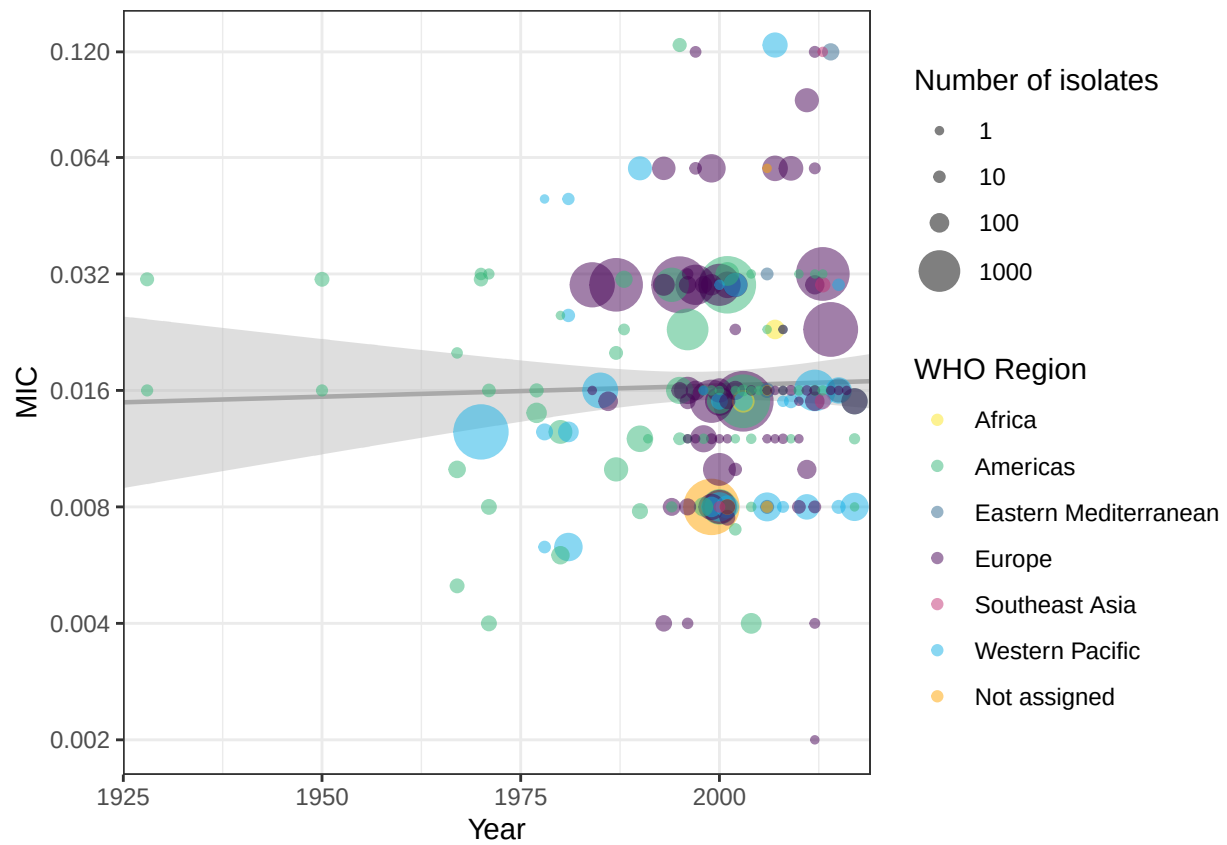
yticks <- c(0, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.12)

p <- ggplot(dat %>%
  filter(!is.na(mic_mix)) %>%
  group_by(isolate_yr, mic_mix, who_name) %>%
  summarise(count = sum(count)) %>%
  mutate(who_name = factor(who_name, levels = c("Africa",
                                                "Americas",
                                                "Eastern Mediterranean",
                                                "Europe",
                                                "Southeast Asia",
                                                "Western Pacific",
                                                "Not assigned")))) %>%
  arrange(desc(count))) +
  geom_ribbon(data = ci, aes(ymin = 2**lwr, ymax = 2**upr, x = isolate_yr),
    alpha = 0.5, fill = "grey", colour = NA) +
  geom_line(data = ci, aes(x = isolate_yr, y = 2**fit),
    colour = "darkgrey", linewidth = 0.8) +
  geom_point(aes(x = isolate_yr, y = mic_mix,
    col = who_name,
    size = count), alpha = 0.5) +
  scale_color_discrete(name = "WHO Region", type = pal) +
  scale_size_continuous(range = c(1,10), name = "Number of isolates", breaks = c(1, 10, 100, 1000)) +
  scale_y_continuous(breaks = yticks, minor_breaks = NULL, name = "MIC") +
  scale_x_continuous(limits = c(X_LOWER, 2017 + 2), expand = c(0,0)) +
  labs(x = "Year", ylab = "MIC") +
  coord_trans(y = "log2") + #, ylim = c(Y_LOWER, 0.14), expand = FALSE) +
  theme_bw()

## `summarise()` has grouped output by 'isolate_yr', 'mic_mix'. You can override
## using the `.groups` argument.

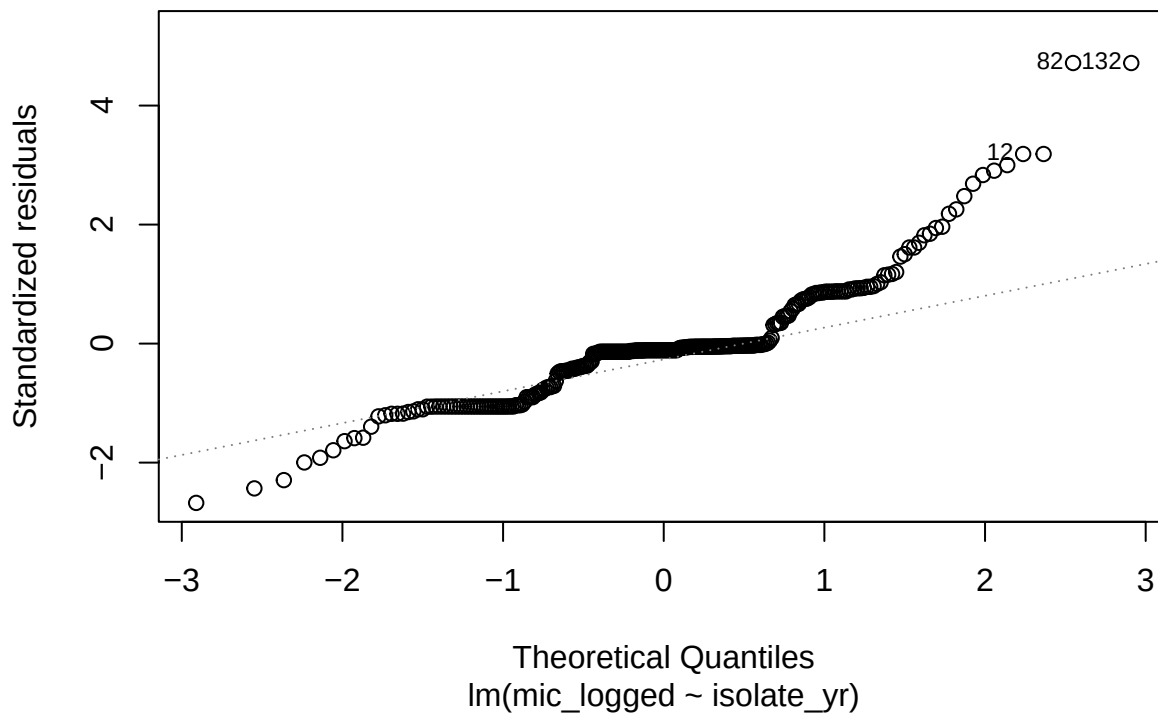
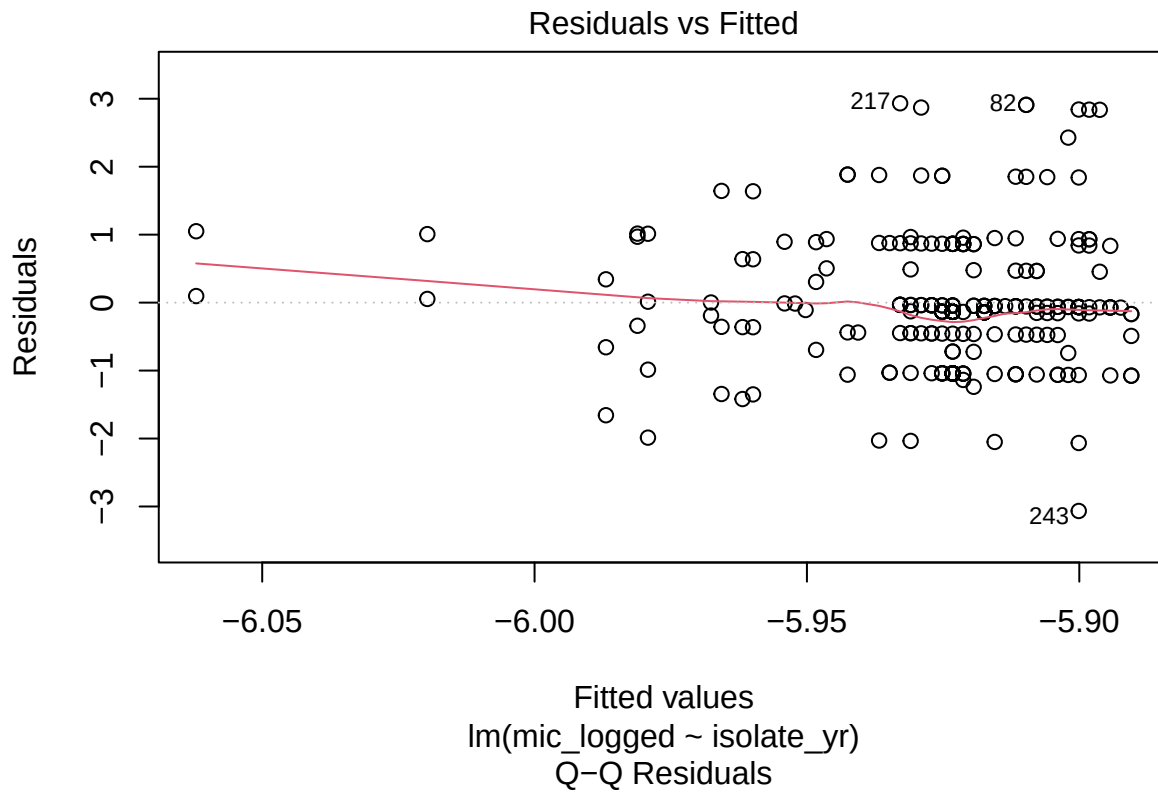
p

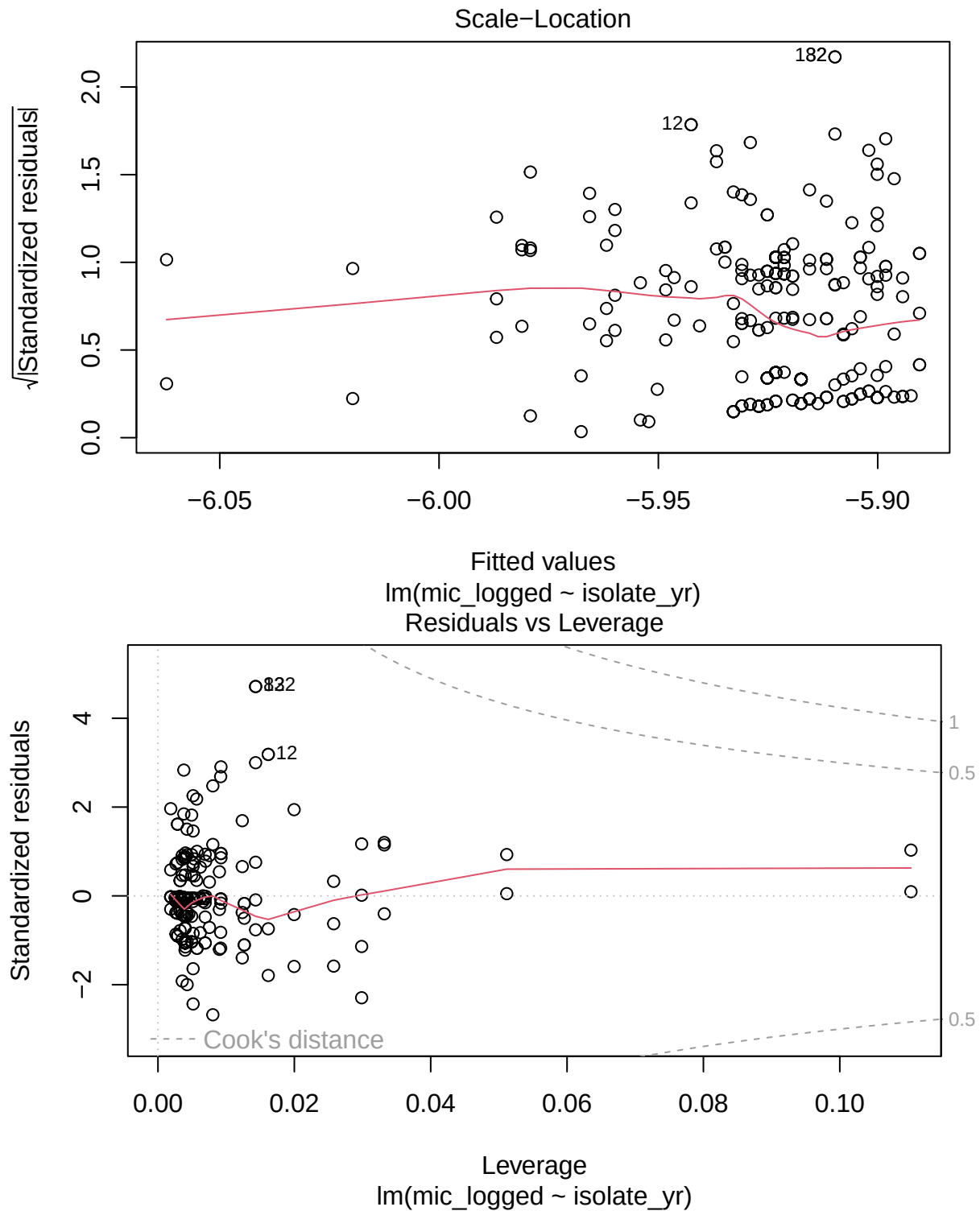
```



```
ggsave("figs/mic_mixed_bubbles_all_logged.png",
       height = 6, width = 9, scale = 0.8)

plot(mod_weighted)
```





Ordinal regression We could really treat MICs as categories as intermediate MICs are not (usually) tested. For this reason, we've had a look at ordinal regression in the chunk below.

```
library(MASS)
```

```
##
```

```

## Attaching package: 'MASS'

## The following object is masked from 'package:dplyr':
##
##      select

fable(dat$mic_mix)

##  0.002 0.004 0.005 0.006 0.0063 0.007 0.0075 0.0078 0.008 0.01 0.012 0.0125 0.014 0.015 0.016 0.02 0
##
##      1      5      1      1      2      1      1      1      49      6      19      4      1      57      59      2
# say we bin MIC ..?
bins <- c(0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.12)
dat$mic_mix_binned <- cut(dat$mic_mix,
                          breaks = c(0, bins, max(dat$mic_mix)),
                          ordered_result = TRUE)
# I've totally forgotten how we got to MIC90s and MICNosses so this ^ may not be
# appropriate

fable(dat$mic_mix_binned)

##  (0,0.002] (0.002,0.004] (0.004,0.008] (0.008,0.016] (0.016,0.032] (0.032,0.064] (0.064,0.12] (0.12,0
##
##      1          5          56          146          49          12          5
# Fit the model - no need to log: that would be for visualisation purposes only under ordinal model
# tried using wts from logged model ..... not thinking too hard about that but we should at some po
# it looks like weights need to be integers as we end up assuming binomial glm for ordinal *logistic* r
# haven't thought too hard about this ... may need to be probit link
model <- polr(mic_mix_binned ~ isolate_yr, data = dat, Hess = TRUE)

summary(model)

## Call:
## polr(formula = mic_mix_binned ~ isolate_yr, data = dat, Hess = TRUE)
##
## Coefficients:
##              Value Std. Error t value
## isolate_yr -0.01635  0.0002099  -77.88
##
## Intercepts:
##              Value      Std. Error t value
## (0,0.002] | (0.002,0.004]   -38.1391      0.0059 -6453.4792
## (0.002,0.004] | (0.004,0.008] -36.5784      0.0318 -1150.3705
## (0.004,0.008] | (0.008,0.016] -33.9443      0.3947  -85.9931
## (0.008,0.016] | (0.016,0.032] -31.5681      0.4197  -75.2133
## (0.016,0.032] | (0.032,0.064] -30.1144      0.4694  -64.1617
## (0.032,0.064] | (0.064,0.12]  -29.1444      0.5405  -53.9167
## (0.064,0.12] | (0.12,0.125]  -28.1691      0.6902  -40.8135
##
## Residual Deviance: 729.6694
## AIC: 745.6694

# note the slope here

new_data = data.frame(isolate_yr = seq(X_LOWER, max(dat$isolate_yr) + 2))

```

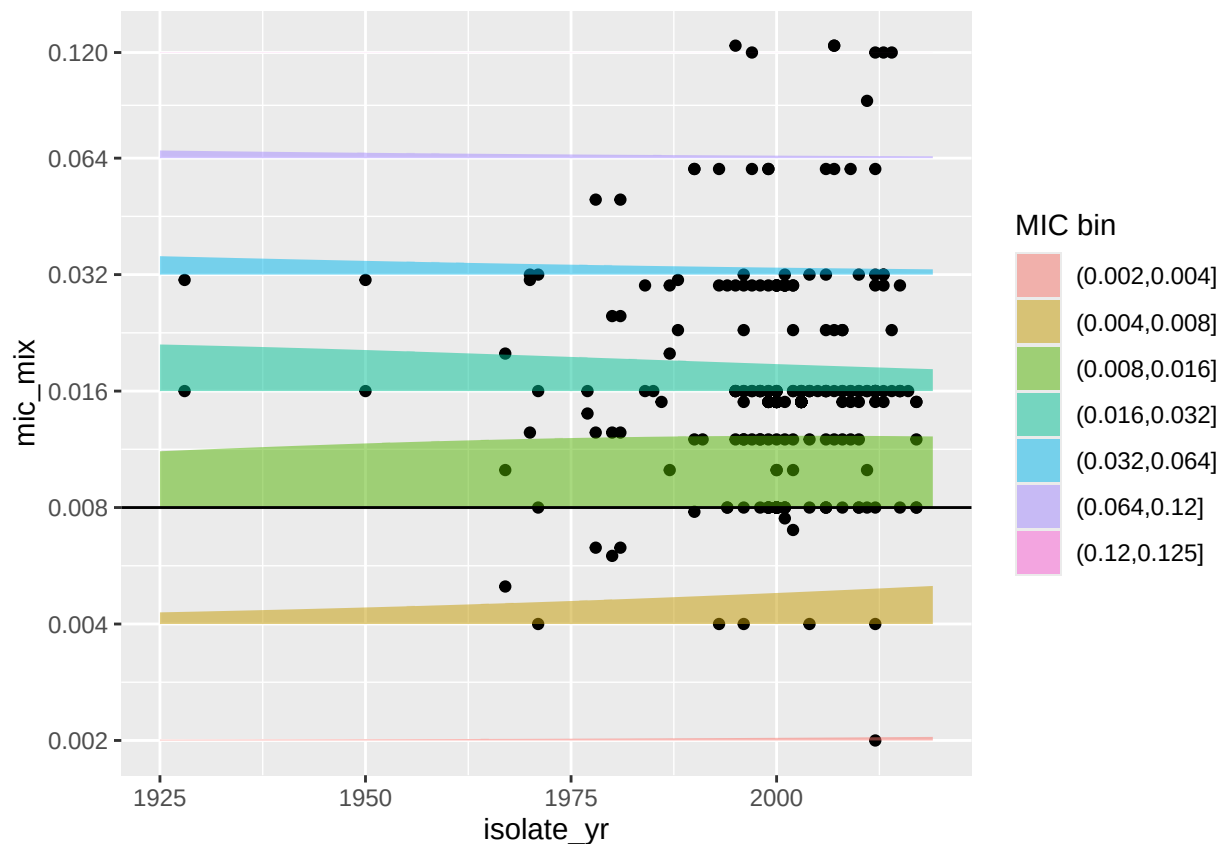
```

preds <- predict(model,
                  newdata = new_data,
                  type = "probs") %>%
  as.data.frame() %>%
  mutate(max_probs = max.col(across(starts_with("(")))) %>%
  mutate(isolate_yr = new_data(isolate_yr))

preds_long <- pivot_longer(preds, starts_with("(")) %>%
  mutate(lower = str_extract(name, "(?<=\\()[^,]+") %>% as.numeric(),
         upper = str_extract(name, "(?<=,)[^]]+") %>% as.numeric(),
         height = lower + (upper - lower)*value) %>%
  filter(lower != 0)

# so now we have colour bars representing class probability between the lower
# and upper extents of the bins ... which don't change much because there isn't
# much change in MIC slope over time
ggplot(data = dat) +
  geom_point(aes(x = isolate_yr, y = mic_mix)) +
  scale_y_continuous(trans = "log2", breaks = bins) +
  geom_ribbon(data = preds_long, aes(ymin = lower, ymax = height, fill = name, x = isolate_yr),
            alpha = 0.5) +
  scale_fill_discrete("MIC bin") +
  # and this black line is the predicted class, which stays the same for the duration of the data
  # (the -1 is because I removed the bottom bin)
  geom_hline(yintercept = bins[preds$max_probs[1] - 1])

```



In the end, we didn't go with ordinal regression, as the class with the greatest probability did not change

over time! And the linear regression gave us a similar result!